Additive Models for Quantile Regression

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Motivation/Provocation

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- Sometimes we are primarily interested in covariate effects on one tail of the conditional distribution of the response.
- Models for conditional quantiles are well-suited to such tasks and easy to estimate via convex optimization.
- Additive modeling for conditional quantiles provides a flexible framework for non-parametric and semi-parametric settings.

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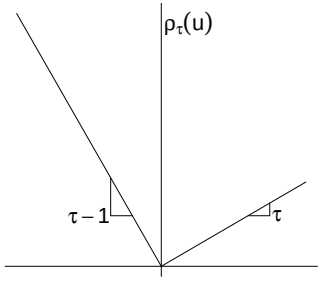
- Quantiles minimize asymmetric linear loss
  - Sorting can be replaced by convex optimization.
- Regression Quantiles also minimize asymmetric linear loss
  - Optimization generalizes nicely to the regression setting, unlike sorting.

Sample Quantiles via Optimization

The τth sample quantile can be defined as any solution to:

$$\hat{\alpha}(\tau) = \text{argmin}_{a \in \Re} \sum_{i=1}^{n} \rho_\tau(y_i - a)$$

where $\rho_\tau(u) = (\tau - I(u < 0))u$ as illustrated below.



Biases the argmin toward making the lower cost error; e.g. forecasting flood levels.

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Similarly, the unconditional τth quantile solves

$$\alpha_\tau = \text{argmin}_a E\rho_\tau(Y - a)$$

and the conditional τth quantile solves

$$\alpha_\tau(x) = \text{argmin}_q E_{Y|X=x}\rho_\tau(Y - q(x))$$

Parametric Quantile Regression

Linear parametric models are simplest:

$$Q_Y(\tau|x) = q(x) = x^\top \beta(\tau)$$

estimable by solving the linear program:

$$\hat{\beta}(\tau) = \text{argmin}_b \sum_{i=1}^{n} \rho_\tau(y_i - x_i^\top b)$$

Solutions have p zero residuals when $\beta \in \mathbb{R}^p$.

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Nonlinear (in parameters) models can also be estimated:

$$\hat{\beta}(\tau) = \text{argmin}_b \sum_{i=1}^{n} \rho_\tau(y_i - g(x_i, b))$$

for some fully specified function, g.

Nonparametric Quantile Regression

Locally Polynomial (Kernel) Method

$$\begin{aligned}\hat{\alpha}(\tau, x) &= \text{argmin}_{\alpha} \sum_{i=1}^{n} \rho_\tau(y_i - \alpha_0 - \alpha_1(x_i - x) - \ldots - \frac{1}{p!}\alpha_p(x_i - x)^p) \\ \hat{g}(\tau, x) &= \hat{\alpha}_0(\tau, x)\end{aligned}$$

Series Methods

$$\begin{aligned}\hat{\alpha}(\tau) &= \text{argmin}_{\alpha} \sum_{i=1}^{n} \rho_\tau(y_i - \sum_{j} \varphi_j(x_i)\alpha_j) \\ \hat{g}(\tau, x) &= \sum_{j=1}^{p} \varphi_j(x)\hat{\alpha}_j\end{aligned}$$

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Penalty Methods

$$\hat{g}(\tau, x) = \text{argmin}_{g} \sum_{i=1}^{n} \rho_\tau(y_i - g(x_i)) + \lambda P(g)$$

Total Variation Regularization I

There are many possible penalties, but total variation of the first derivative of g is particularly attractive:

$$P(g) = V(g') = \int |g''(x)| dx$$

As $\lambda \to \infty$ we constrain g to be closer to linear in x. Solutions of

$$\min_{g \in \mathcal{G}} \sum_{i=1}^{n} \rho_\tau(y_i - g(x_i)) + \lambda V(g')$$

are continuous and piecewise linear.

Total Variation Regularization II

For bivariate functions we consider the analogous problem:

$$\min_{g \in \mathcal{G}} \sum_{i=1}^{n} \rho_\tau(y_i - g(x_{1i}, x_{2i})) + \lambda V(\nabla g)$$

where the total variation variation penalty is now:

$$V(\nabla g) = \int \|\nabla^2 g(x)\| dx$$

Solutions are again continuous, but now they are piecewise linear on a triangulation of the observed x observations. Again, as $\lambda \to \infty$ solutions are forced toward linearity.

Additive Models: Putting it all together

We can combine such models:

$$\min_{g \in \mathcal{G}} \sum_{i=1}^{n} \rho_\tau(y_i - \sum_j g_j(x_{ij})) + \sum_j \lambda_j V(\nabla g_j)$$

- Components g_j can be univariate, or bivariate.
- Additivity is intended to muffle the curse of dimensionality.
- Linear terms are easily allowed, or enforced.
- And shape restrictions like monotonicity and convexity/concavity as well as boundry conditions on g_j's can also be imposed.

Implementation in the R quantreg Package

- Problems are typically large, very sparse linear programs.
- Optimization via interior point methods are quite efficient,
- Provided sparsity of the linear algebra is exploited, quite large problems can be estimated.
- The nonparametric qss components can be either univariate, or bivariate
- Each qss component has its own λ specified
- Linear covariate terms enter formula in the usual way
- The qss components can be shape constrained.

```
fit <- rqss(y ~ qss(x1,3) + qss(x2,8) + x3, tau = .6)
```

Example 1: Fish in a Bottle

Objective: to study metabolic activity of various fish species in an effort to better understand the nature of the feeding cycle. Metabolic rates based on oxygen consumption as measured by sensors mounted on the tubes.

Three primary aspects are of interest:

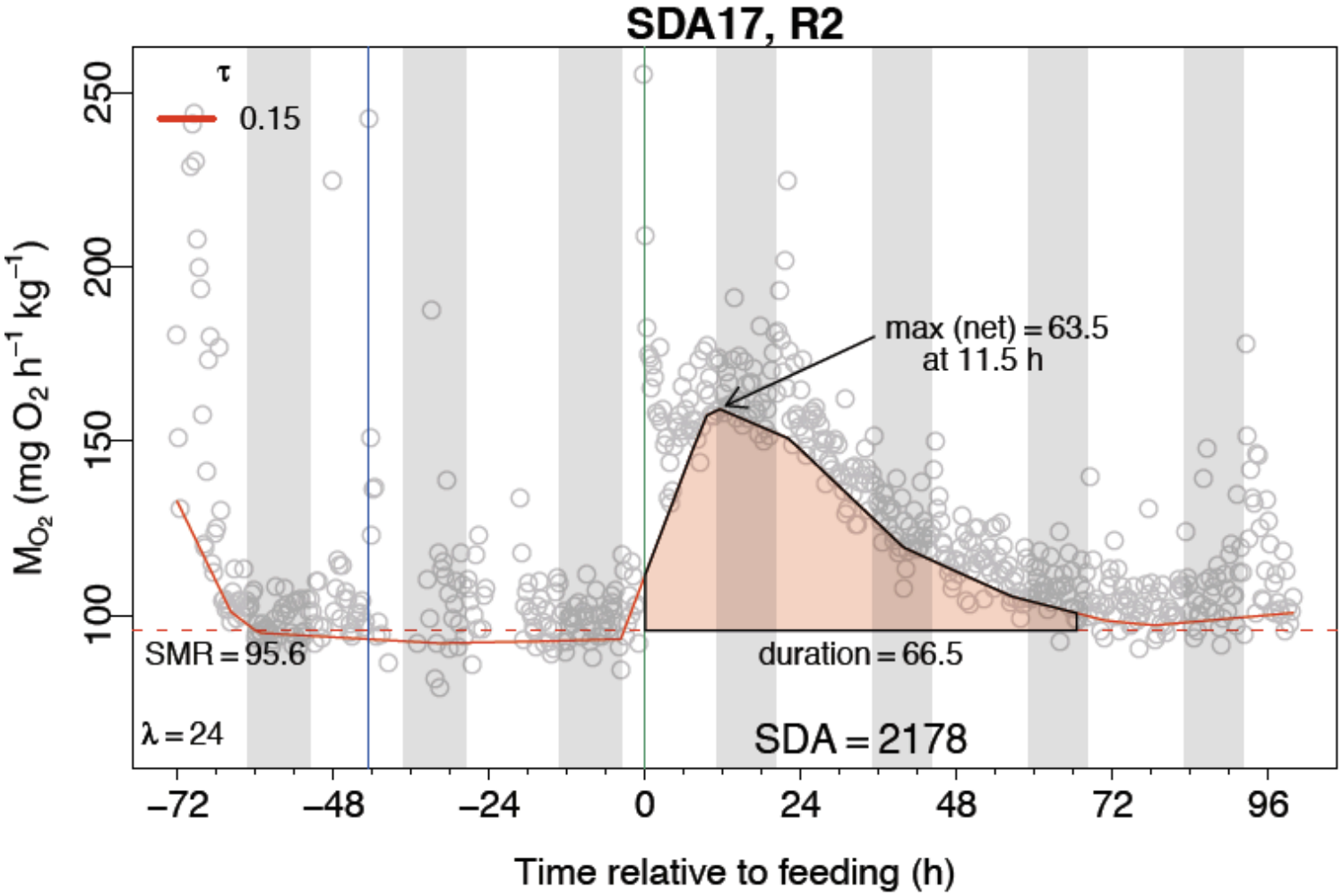
1. Basal (minimal) Metabolic Rate,
2. Duration and Shape of the Feeding Cycle, and
3. Diurnal Cycle.

Example 1: Some Experimental Details

Experimental data of Denis Chabot, Institut Maurice-Lamontagne, Quebec, Canada and his colleagues.

1. Basal (minimal) metabolic rate $\mathrm{M}_{\mathrm{O}_2}$ (aka Standard Metabolic Rate SMR) is measured in $\mathrm{mg\ O_2\ h^{-1}\ kg^{-1}}$ for fish "at rest" after several days without feeding,
2. Fish are then fed and oxygen consumption monitored until $\mathrm{M}_{\mathrm{O}_2}$ returns to its prior SMR level for several hours.
3. Elevation of $\mathrm{M}_{\mathrm{O}_2}$ after feeding (aka Specific Dynamic Action SDA) ideally measures the energy required for digestion,
4. Procedure is repeated for several cycles, so each estimation of the cycle is based on a few hundred observations.

Example 1: Juvenile Codfish



Tuning Parameter Selection

There are two tuning parameters:

1. $\tau = 0.15$ the (low) quantile chosen to represent the SMR,
2. λ controls the smoothness of the SDA cycle.

One way to interpret the parameter λ is to note that it controls the number of effective parameters of the fitted model.

$$p(\lambda) = \|\hat{\beta}(\lambda)\|_0 = \text{card}\{i : \hat{\beta}_i(\lambda) \neq 0\}$$

This is equivalent to the number of interpolated observations, the number of zero residuals.

$$p(\lambda) = \text{div } \hat{g}_{\lambda,\tau}(y_1, ..., y_n) = \sum_{i=1}^{n} \partial\hat{y}_i/\partial y_i$$

Example 2: Childhood Malnutrition in India

A larger scale problem illustrating the use of these methods is a model of childhood malnutrition considered by Fenske, Kneib and Hothorn (2009).

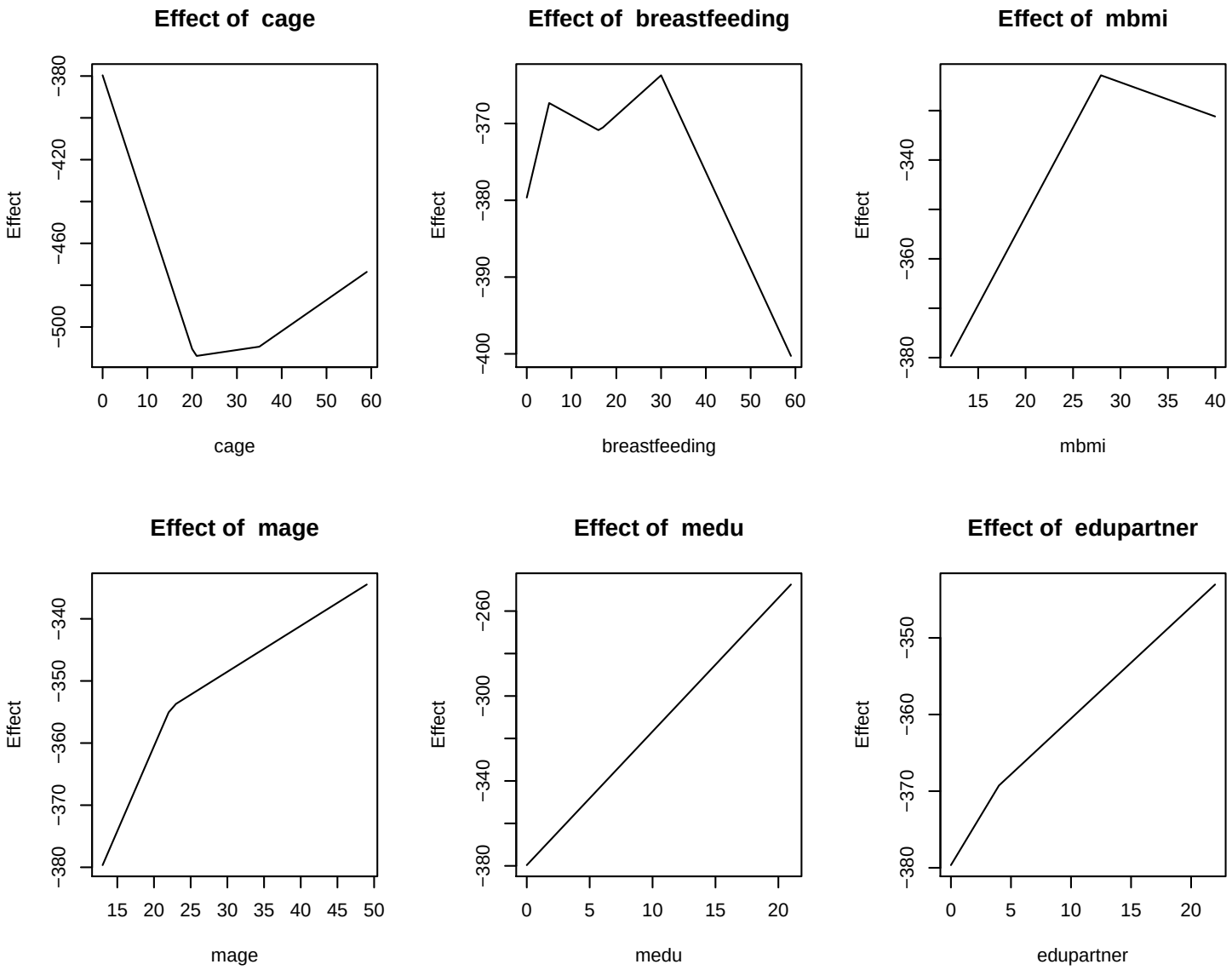
- They motivate the use of models for low conditional quantiles of height as a way to explore influences on malnutrition,
- They employ boosting as a model selection device,
- Their model includes six univariate nonparametric components and 15 other linear covariates.
- There are 37,623 observations on children's heights

Example 2: R Formulation

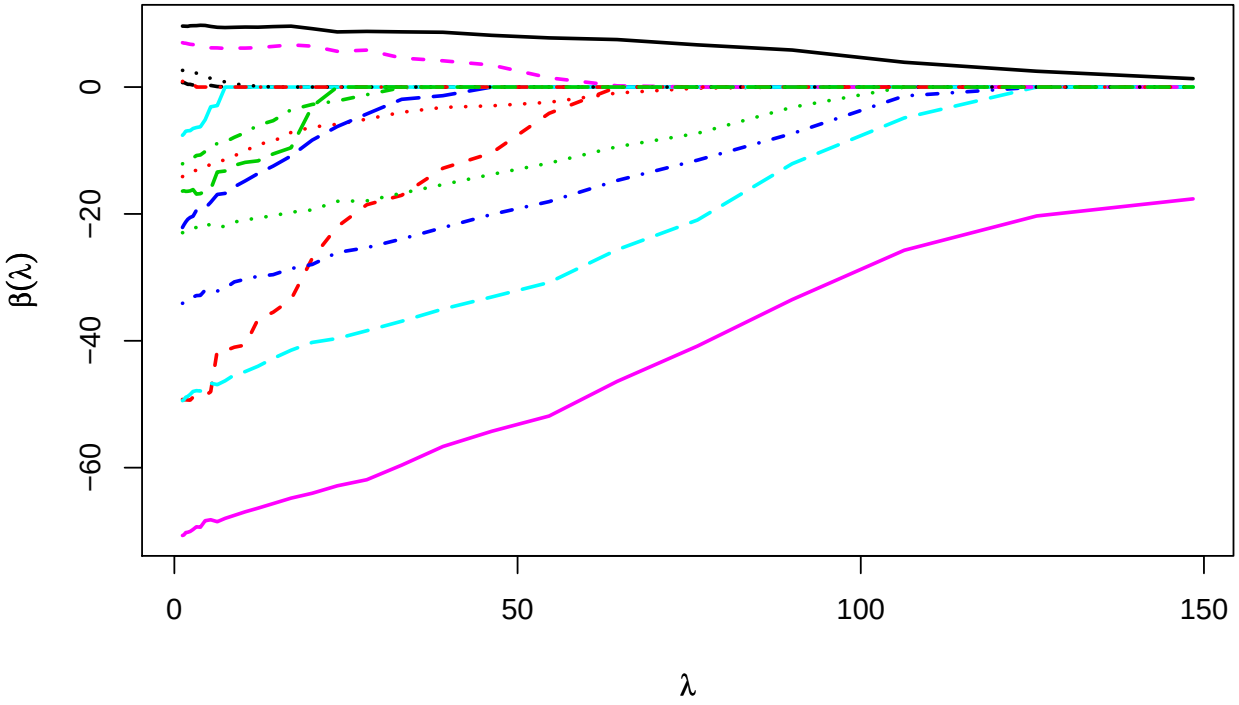
```
rqss(stunting ~ csex + ctwin + cbirthorder + munemployed +
mreligion + mresidence + deadchildren + wealth + electricity +
radio + television + frig + bicycle + motorcycle + car +
qss(cage, lambda = lamss[1]) + qss(bfed, lambda = lamss[2]) +
qss(mbmi, lambda = lamss[3]) + qss(mage, lambda = lamss[4]) +
qss(medu, lambda = lamss[5]) + qss(fedu, lambda = lamss[6]) +
tau = 0.10, method = "lasso", lambda = lambda, data = india)
```

- The six coordinates of `lamss` control the smoothness of the nonparametric components,
- `lambda` controls the degree of shrinkage in the linear (lasso) coefficients.
- The estimated model has roughly 40,000 observations, including the penalty contribution, and has 2201 parameters.
- Fitting the model for a single choice of λ's takes approximately 5 seconds.

Example 2: Nonparametric Components



Example 2: Lasso Shrinkage of Linear Components



Of the 15 original covariates that were introduced linearly, the lasso selection with λ chosen to be 100 selects four: the gender of the child, whether the mother is employed, her religion, and whether she is "urban" or "rural."

Conclusions

- Quantile regression provides a unified approach to the estimation of conditional quantile functions just as least squares and related robust methods estimate models for conditional central tendency.
- Total variation roughness penalties offer an attractive approach to nonparametric quantile regression.
- Additive models are easily implemented combining nonparametric and linear parametric components.
- Conventional lasso penalties can be used as a model selection device for linear components.
- AIC-like methods are available for choosing penalty parameters based on simple measures of the effective dimensionality of the fitted models.